Identifying Monetary Policy Shocks in an Administered-Rate Economy: Evidence from Peru

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Abstract

We compare seven identification strategies for monetary policy transmission in Peru and find that only recursive (Cholesky) identification produces interpretable estimates—modern methods fail. Three institutional features explain this: the interbank rate is administered by construction, no rate-futures market exists for policy surprise extraction, and the monetary record provides too few sharp episodes for narrative restrictions. The Cholesky VAR(1) yields a peak GDP response of -0.195 pp per 100 bp (90% CI: $[-0.698, +0.271]$), consistent with the Peru SVAR literature. Our most robust result is a GDP–poverty reduced-form coefficient of $\hat{\beta} = -0.656$ ($t = -5.69$, $R^2 = 0.67$, $N = 18$), stable across sub-periods. Chaining the two estimates implies a 100 bp hike may raise poverty by approximately $+0.13$ pp. These three features characterize a broad class of EME central banks, making identification methodology—not just Peru's results—the central contribution.

JEL codes: E52, E32, I32, C32.
Keywords: monetary policy transmission, SVAR, local projections, narrative identification, Proxy-SVAR, poverty, Peru, emerging markets.

1 Introduction

Understanding how monetary policy transmits to real activity is a central question in macroeconomics. The global literature has developed an expanding identification toolkit: recursive (Cholesky) ordering, sign restrictions (Uhlig, 2005), narrative restrictions (Antolín-Díaz and Rubio-Ramírez, 2018), and external-instrument Proxy-SVARs (Stock and Watson, 2018; Gertler and Karadi, 2015; Montiel Olea et al., 2021). These methods have been applied extensively to the United States and Euro Area, and increasingly to advanced economies such as Australia. Their applicability to small open emerging-market economies (EMEs) with institutional features that differ markedly from the US—administered policy rates, no OIS market, limited monetary history—has received far less systematic attention.

Peru offers a particularly instructive case. The Banco Central de Reserva del Perú (BCRP) has operated a credible inflation-targeting framework since 2002, with a target of $2\% \pm 1\,\text{pp}$. Yet Peru's monetary implementation differs fundamentally from the US Federal Reserve: the interbank overnight rate is *administered*—the BCRP ensures the interbank rate equals the reference rate by construction, leaving no market-determined component from which to extract policy surprises. This single institutional feature, combined with the absence of rate-futures markets and a relatively uneventful monetary history, renders the modern identification toolkit mostly inapplicable.

The existing Peru-specific literature—Pérez Rojo and Rodríguez (2024), Castillo et al. (2011), and Portilla et al. (2022)—uniformly relies on recursive identification without documenting why more modern alternatives fail. This paper fills that gap with four contributions:

1. **First systematic comparison** of seven identification strategies for monetary policy in Peru within a single unified dataset ($T = 85$, 2004Q2–2025Q3; see Table 12).

2. **A cautionary result for the communication-as-instrument literature:** our LLM-classified BCRP tone instrument (325 *Notas Informativas*, 2001–2026; Table 3) passes relevance only when it retains anticipated policy content, violating exogeneity

and producing wrong-signed LP-IV responses at all horizons $h = 0$–12. This documents the anticipated-policy problem in a central bank whose communication closely tracks its decisions.

3. **A well-identified GDP–poverty reduced-form coefficient** ($\hat{\beta} = -0.656$, $p < 0.001$, $N = 18$; Table 10) that chains monetary policy effects to poverty outcomes, stable across sub-periods (Table 11).

4. **An explanation of why modern identification fails** in Peru's institutional environment, contributing to the emerging literature on monetary policy transmission in EMEs (Ha et al., 2025).

Beyond the Peru-specific results, this paper makes a methodological contribution to the EME literature. The three features that block modern identification in Peru—an administered interbank rate, the absence of rate-futures markets, and a limited number of sharp monetary episodes—are not idiosyncratic to Peru. They characterize a large class of EME central banks spanning Latin America, the Middle East, and South Asia (Table 13). For this class, the recursive SVAR is not a fallback for lack of sophistication; it is the credible identification frontier. Understanding *why* modern methods fail in these settings, and what diagnostic tests reveal those failures, is the methodological contribution of this paper.

The Cholesky benchmark (Figure 3) yields a peak GDP response of -0.195 pp per 100 bp consistent with published Peru estimates. Sign restrictions produce an identified set too wide for inference. Narrative restrictions are blocked by a fundamental COVID-dummy–narrative tension. Proxy-SVAR with the interbank rate fails relevance ($F = 4.73$; Table 8), and Proxy-SVAR with tone passes relevance only when the instrument retains anticipated policy (violating exogeneity; Table 9).

Section 2 describes Peru's monetary framework. Section 3 presents the data. Section 4 develops the econometric framework. Section 5 reports results (Section 5.9 provides the systematic comparison). Section 6 discusses implications. Section 7 concludes.

2 Institutional Background

2.1 BCRP Monetary Policy Framework

The BCRP adopted inflation targeting in 2002, with a target band of $2\% \pm 1$ pp. The reference rate is announced monthly on a pre-set calendar. Unlike the Federal Reserve, the BCRP keeps the interbank overnight rate equal to the reference rate through active liquidity provision or absorption, making it administered by construction.

Additional distinguishing features include: (i) active foreign exchange intervention in a managed float; (ii) partial dollarization, with significant deposits and credit in both Peruvian soles (PEN) and US dollars; and (iii) the absence of any OIS or rate-futures market for PEN, precluding high-frequency extraction of market rate expectations.

2.2 Why Peru Differs from the US and Euro Area

Three features make modern identification strategies particularly difficult.

Administered interbank rate. The interbank rate equals the reference rate by BCRP design, leaving no "surprise" component in the sense of Gertler and Karadi (2015). The interbank "surprise" used as a Proxy-SVAR instrument is mechanically collinear with the reference rate change already in the VAR (Section 5.5).

No rate expectations market. Without OIS or futures contracts on PEN rates, it is impossible to separate the expected from the unexpected component of a rate decision, as is standard in the US literature (Miranda-Agrippino and Ricco, 2021).

Limited sharp monetary episodes. Since 2002, BCRP changes have been small and gradual, with only two exceptions: the emergency COVID cuts of 2020 (2.25% to 0.25%) and the 2021–2022 hiking cycle (0.25% to 7.75%). As Read and Zubairy (2023) demonstrate,

even the US narrative identification literature struggles with fewer than a handful of clear episodes; two episodes are insufficient for Peru.

2.3 BCRP Reference Rate History

Figure 1 shows the BCRP reference rate from 2003 to 2025Q3, with key episodes annotated: the GFC cutting cycle (2008–2009, from 6.5% to 1.25%), the 2010–2011 hiking cycle (to 4.25%), gradual cuts in 2014–2015, the COVID emergency cuts to 0.25% in 2020, the aggressive 2021–2022 hiking cycle to 7.75%, and subsequent easing to 4.25% by end-2025.

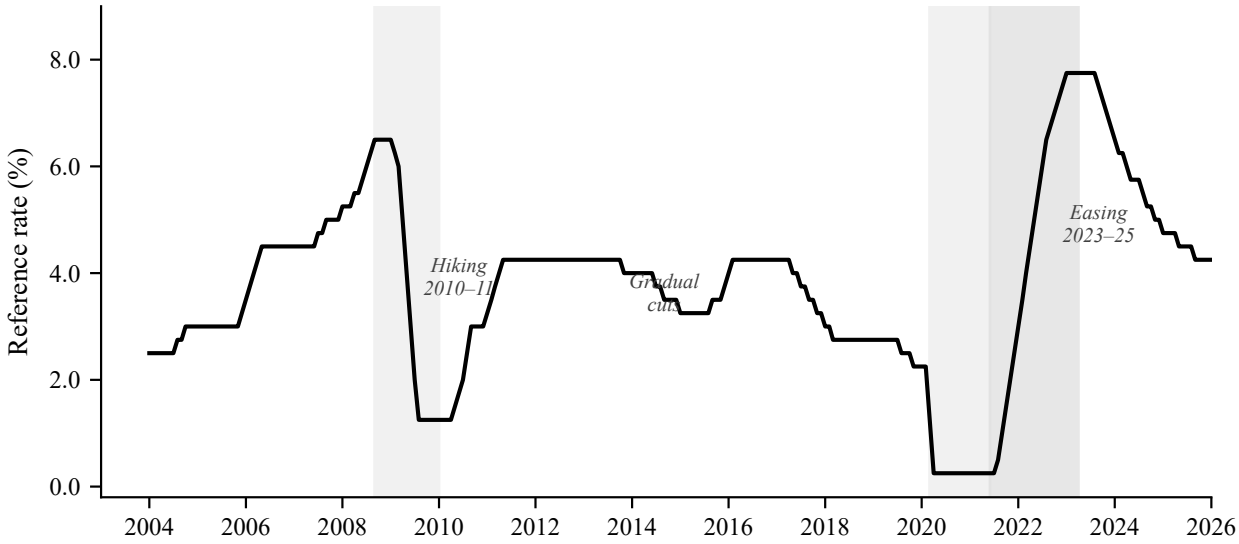

Figure 1: BCRP Reference Rate, 2003–2025. Key episodes: GFC cuts (2008–2009), 2010–2011 hiking cycle, COVID emergency cuts to 0.25% (2020), aggressive hiking to 7.75% (2021–2022), and gradual easing (2023–2025). Source: BCRP series PD04722MM.

3 Data

3.1 Macroeconomic Variables

The VAR uses five quarterly series from the BCRP data portal: terms of trade (PN38923BM), real GDP quarter-on-quarter growth (PN01731AM, summed from monthly seasonally adjusted index), CPI quarter-on-quarter change (PN01271PM), PEN/USD exchange rate change

(PN01246PM), and BCRP reference rate change (PD04722MM). The sample runs from 2004Q2 to 2025Q3 ($T = 85$). Summary statistics are in Table 1; all series are stationary by ADF tests ($p < 0.001$ for each).

COVID treatment follows the Frisch-Waugh-Lovell (FWL) principle: indicator variables for 2020Q1 and 2020Q2 are partialled out of all series before estimation. This is preferred to a contemporaneous dummy in the VAR because a dummy in Y_t but not the lags creates inconsistency in VAR($p \geq 2$) systems by contaminating lagged regressors.

3.2 BCRP Communications Corpus

We construct a dataset of 325 *Notas Informativas del Programa Monetario*—the official monthly communications explaining each rate decision—covering 2001 to 2026. Documents were downloaded from the BCRP website and the Wayback Machine; text was extracted with `pdfplumber`.

Each note is classified by tone using two methods: (a) a dictionary-based approach using a hawkish/dovish lexicon adapted from Lahura and Vega (2020), producing scores in $[-1, +1]$; and (b) an LLM-based approach using Claude Haiku, prompted to score each note on $[-100, +100]$. Table 3 presents classification statistics. The correlation between dictionary and LLM scores is 0.373, indicating moderate agreement. Both methods correctly flag 2009 as dovish and 2022 as hawkish. Figure 2 shows the tone series over time.

Year-by-year statistics for both methods appear in Table 4 (Appendix C).

3.3 Poverty Data

The poverty measure is the official annual monetary poverty headcount ratio from INEI/ENAHO household surveys, 2005–2024. COVID years 2020–2021 are excluded due to lockdown disruptions and INEI methodology changes. The regression sample contains $N = 18$ annual observations. Table 2 lists the full dataset.

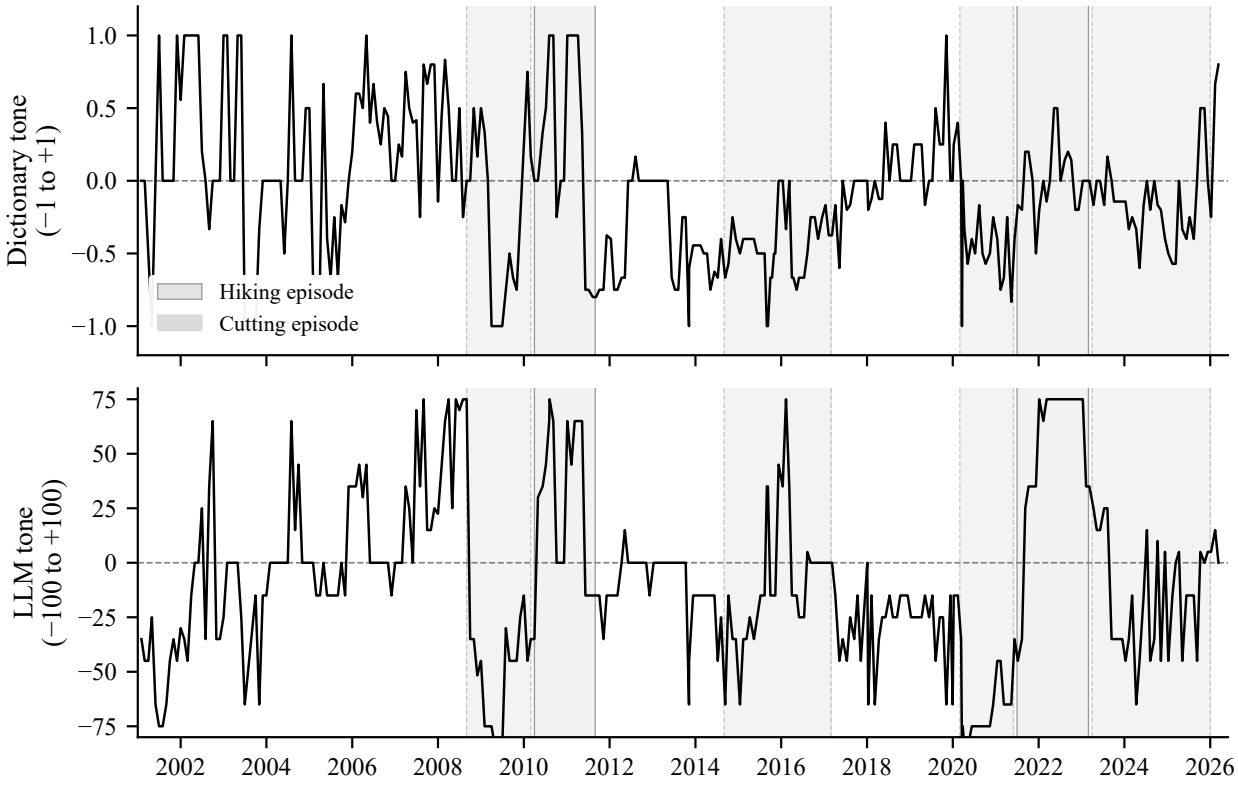

Figure 2: BCRP Communication Tone, 2001–2026. *Top panel:* dictionary score (−1 to +1). *Bottom panel:* LLM score (−100 to +100). Hatching: hiking episodes (///); cutting episodes (. . .). Both measures flag 2009 (GFC emergency cuts) as strongly dovish and 2022 (hiking cycle) as hawkish. Dictionary–LLM correlation: $r = 0.373$.

4 Econometric Framework

4.1 Reduced-Form VAR

The baseline model is a VAR(p):

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $Y_t = (\text{tot}_t, \text{gdp}_t, \text{cpi}_t, \text{fx}_t, \text{rate}_t)'$. COVID quarters are handled via FWL. AIC and BIC both select $p = 1$.

4.2 Identification Strategies

Strategy 1 (Cholesky). We impose a lower-triangular Cholesky decomposition $\hat{\Sigma} = PP'$ and identify the monetary policy shock as the innovation to the rate equation after conditioning on same-quarter innovations to ToT, GDP, CPI, and FX. The ordering ToT $\rightarrow$ GDP $\rightarrow$ CPI $\rightarrow$ FX $\rightarrow$ Rate embeds the assumption that the BCRP does not react within-quarter to real activity or inflation, consistent with Peru's monthly-calendar announcement system.

Strategy 2 (Sign Restrictions). Following Uhlig (2005), we identify the monetary shock ε_t^{mp} by requiring

$$\frac{\partial\, \text{rate}_{t+h}}{\partial\, \varepsilon_t^{mp}} > 0 \;\text{ for } h = 0, \qquad \frac{\partial\, \text{gdp}_{t+h}}{\partial\, \varepsilon_t^{mp}} \leq 0 \;\text{ for } h = 1, 2, 3. \tag{2}$$

We use the algorithm of Rubio-Ramírez et al. (2010): draw Q from the Haar measure on $O(n)$ via QR decomposition of a random normal matrix, accept draws satisfying the sign restrictions.

Strategy 3 (Narrative Sign Restrictions). Following Antolín-Díaz and Rubio-Ramírez (2018), we add episode-level restrictions. Letting q identify the monetary policy shock column:

$$q' u_{2020\text{Q}1} < 0 \quad \text{(March 2020: expansionary shock)}, \tag{3}$$

$$q' u_{2022\text{Q}k} > 0 \quad \text{for } k = 1, \ldots, 4 \text{ (2022 hiking cycle: contractionary)}. \tag{4}$$

We test multiple combinations of episode restrictions, with and without the FWL COVID dummy.

Strategy 4 (Local Projections). Following Jordà (2005):

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \, \Delta r_t + \gamma_{1h} \, y_{t-1} + \gamma_{2h} \, y_{t-2} + \delta_{1h} \, \Delta r_{t-1} + \delta_{2h} \, \Delta r_{t-2} + \lambda \, \text{covid}_t + \varepsilon_{t,h}, \tag{5}$$

with HC3 standard errors. Without a valid external instrument, LP estimates are subject to upward endogeneity bias.

Strategy 5 (Proxy-SVAR: Interbank Rate). Following Stock and Watson (2018) and Gertler and Karadi (2015), the instrument is $z_t = \text{interbank}_{d+1} - \text{interbank}_{d-1}$ around announcement date d, aggregated to quarters. Conditions: relevance $E[z_t \varepsilon_t^{mp}] \neq 0$ and exogeneity $E[z_t \varepsilon_t^{j}] = 0$ for $j \neq mp$.

Strategy 6 (Proxy-SVAR: Communication Tone). We construct two tone-surprise instruments: Construction A residualizes tone on Δr_t plus lagged macro (purging anticipated decisions); Construction B residualizes on lagged macro only. We assess both using first-stage F-statistics (Montiel Olea et al., 2021) and LP-IV sign checks.

Strategy 7 (GDP–Poverty OLS).

$$\Delta \text{poverty}_t = \alpha + \beta \cdot \text{GDP_growth}_t + \varepsilon_t, \tag{6}$$

OLS on $N = 18$ annual observations. This is a reduced-form relationship, chained with the Cholesky IRF to compute the full rate $\rightarrow$ GDP $\rightarrow$ poverty transmission.

5 Results

All tables are collected in Appendix A; figures appear inline in the text.

5.1 Cholesky VAR(1): Benchmark

Table 5 reports the estimated $\hat{A}_1$ matrix. The largest own-lag is the reference rate (0.614), reflecting rate smoothing. GDP has a negative own-lag (-0.264), consistent with mean reversion in quarterly growth.

Figure 3 plots impulse responses of GDP, CPI, and the exchange rate to a 100 bp monetary policy shock, with 90% bootstrap confidence bands (2,000 replications). The peak GDP response is -0.195 pp at $h = 3$ quarters (Table 12). The 90% CI is $[-0.698, +0.271]$, including zero—expected given $T = 85$ quarterly observations—and consistent with the existing Peru literature (-0.25 to -0.30 pp; Table 12).

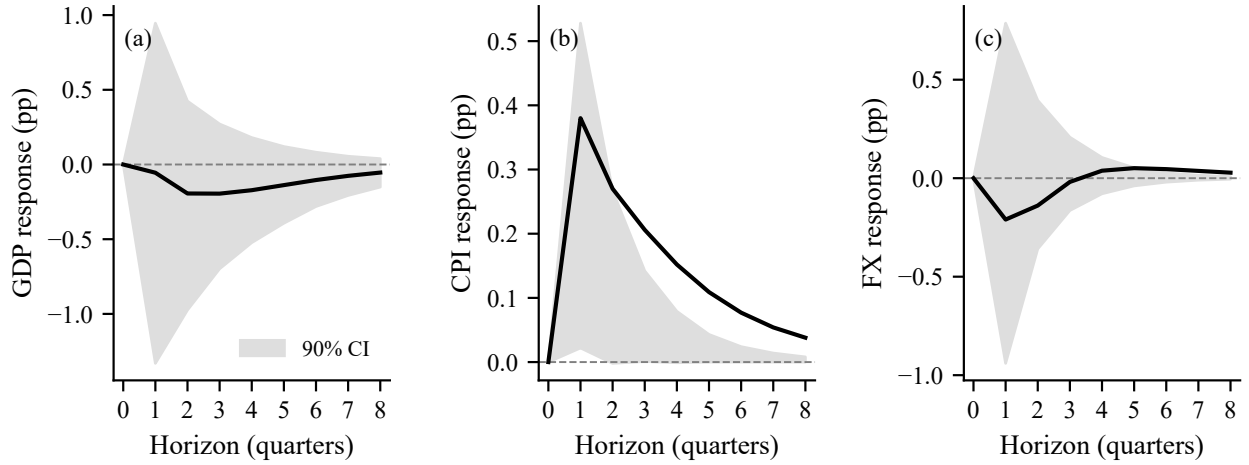

Figure 3: Cholesky VAR(1): Impulse Responses to a 100 bp Rate Shock ($h = 0, \ldots, 8$ quarters). Peak GDP response: -0.195 pp at $h = 3$. Shaded bands: 90% bootstrap CI (2,000 draws). The CI includes zero at all horizons, typical for $T = 85$ quarterly VARs. Ordering: ToT $\rightarrow$ GDP $\rightarrow$ CPI $\rightarrow$ FX $\rightarrow$ Rate.

5.2 Sign Restrictions

Table 6 (first row) and Figure 4 report sign restriction results. Of 80,000 candidate rotation matrices, 63,743 (79.7%) satisfy the restrictions—the restrictions are weak relative to the data. The median peak GDP response is −2.30 pp with a 68% credible set of [−12.8, +5.4] pp. The identified set is too wide to support point inference. Additional restrictions are required.

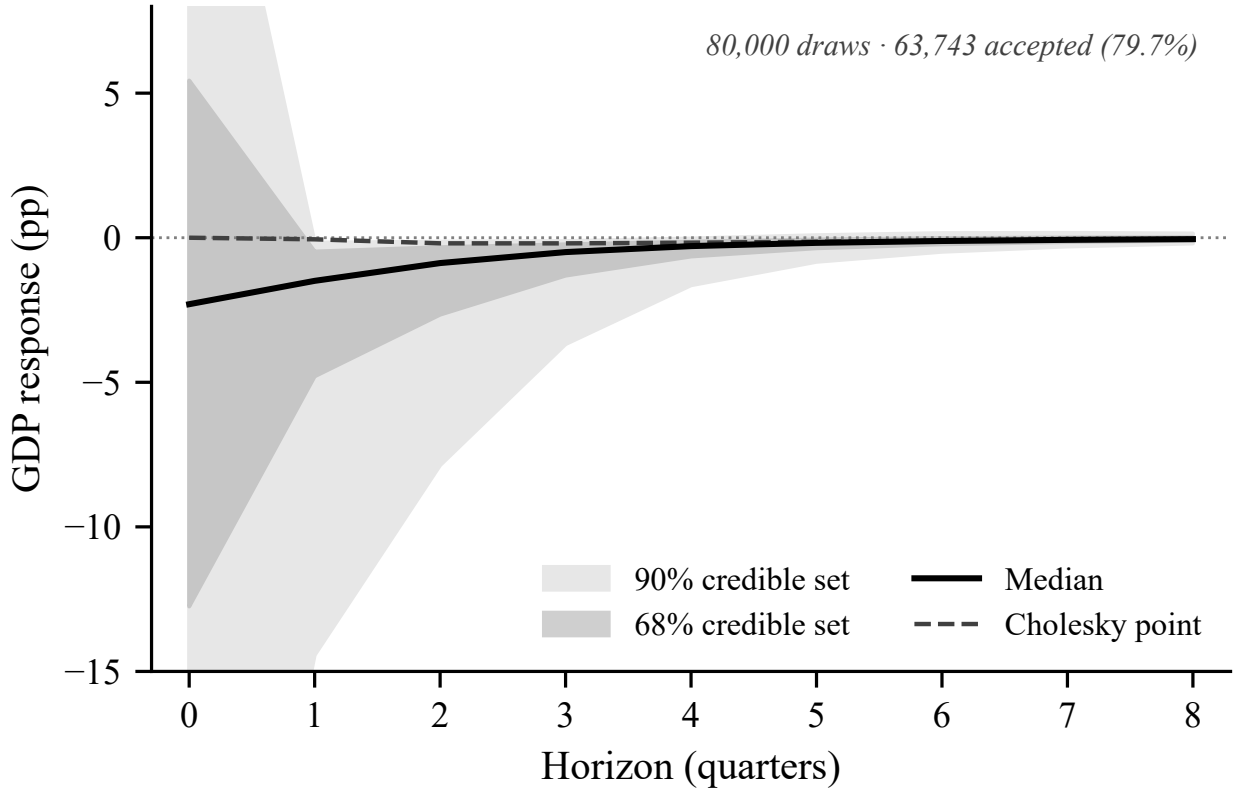

Figure 4: Sign-Restricted Identified Set for the Peak GDP Response. Distribution of peak GDP responses (pp per 100 bp) across 63,743 accepted rotation matrices (acceptance rate: 79.7%). The set is wide and straddles zero, making point estimation infeasible.

5.3 Narrative Sign Restrictions

Table 6 reports acceptance rates and median peak GDP responses across five specifications. The fundamental tension is the following. Applying narrative restrictions on 2020Q1 combined with the FWL COVID dummy yields only ≈ 4 accepted draws of 80,000: the FWL treatment absorbs the variation that narrative restrictions on 2020 quarters need. This is

not a specification error but an inherent conflict between outlier treatment and narrative identification when the narrative event is itself the outlier.

Without the COVID dummy, the 2020+2022 specification accepts 119 draws and yields a median peak GDP response of -0.74 pp (68% CI: $[-1.06, -0.38]$). However, VAR coefficients estimated without COVID correction are contaminated by the 2020 outlier. As Read and Zubairy (2023) demonstrate, single-episode narrative restrictions are insufficient for robust identification even in the US; two episodes are insufficient for Peru.

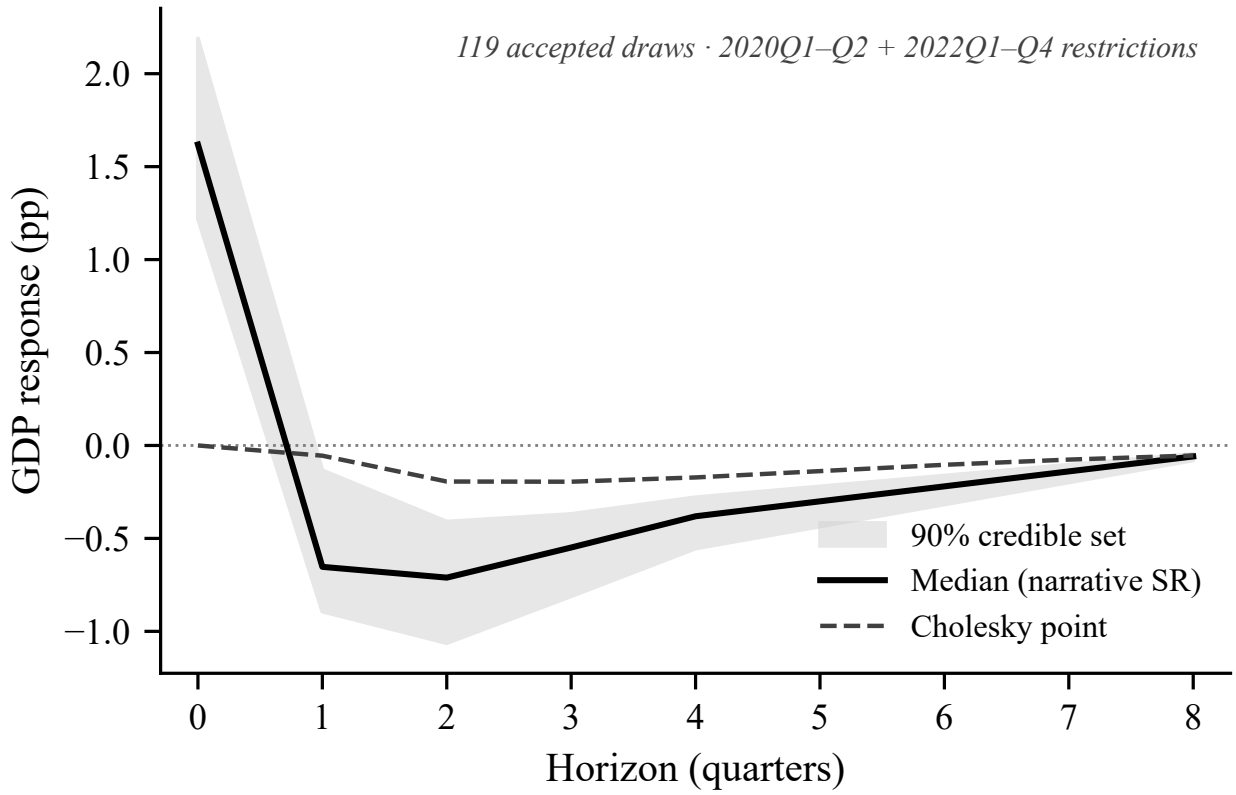

Figure 5: Narrative Sign Restriction IRFs (Best Specification: 2020+2022 episodes, no FWL, 119 accepted draws). Median (solid line), 68% (dark) and 90% (light) credible sets. Peak GDP at -0.74 pp; the interval is very wide, reflecting the sparse identification.

5.4 Local Projections

Table 7 reports LP estimates at horizons $h = 0, \ldots, 8$. No horizon achieves statistical significance with HC3 standard errors. Positive coefficients at $h = 0$ and $h = 1$ reflect the

classic endogeneity problem: the BCRP raises rates when GDP is strong. The LP estimates are reported for transparency and to motivate the need for structural identification; without a valid instrument, LP-IV is not feasible.

5.5 Proxy-SVAR: Interbank Rate

Table 8 reports feasibility diagnostics. Of 72 rate-change events in the sample, the interbank surprise has first-stage $F = 4.73$, well below the $F > 10$ threshold of Stock and Watson (2018). Peru's interbank rate equals the reference rate by construction, so the "surprise" is mechanically collinear with the reference rate change already in the VAR. The relevance condition fails.

5.6 Proxy-SVAR: BCRP Communication Tone

Table 9 reports instrument validity tests for the two constructions. Construction A (tone residualized on Δr_t and lagged macro) has $F = 4.75$: after purging the rate decision, residual tone carries near-zero information about monetary shocks. Construction B (residualized on lagged macro only) has $F = 33.7$, satisfying relevance. However, Figure 6 shows that LP-IV GDP responses are positive at all horizons (+4.32 pp at $h = 0$, +7.36 pp at $h = 1$)—the wrong sign for a contractionary shock. Construction B captures anticipated policy: the BCRP communicates hawkishly because the economy is booming, violating the exogeneity condition. This is a cautionary result for the growing literature on central bank communication as identification (Wolf, 2020): high F does not guarantee exogeneity.

Limitations and alternative specifications. Several caveats attach to the tone instrument result. First, the LLM-dictionary correlation of 0.373 indicates that LLM scores are noisy; results may differ with better-calibrated scoring or with a purpose-built hawkishness lexicon for Spanish-language central bank communications. Second, we have used tone levels throughout; an alternative specification using changes in tone (Δtone) may better isolate

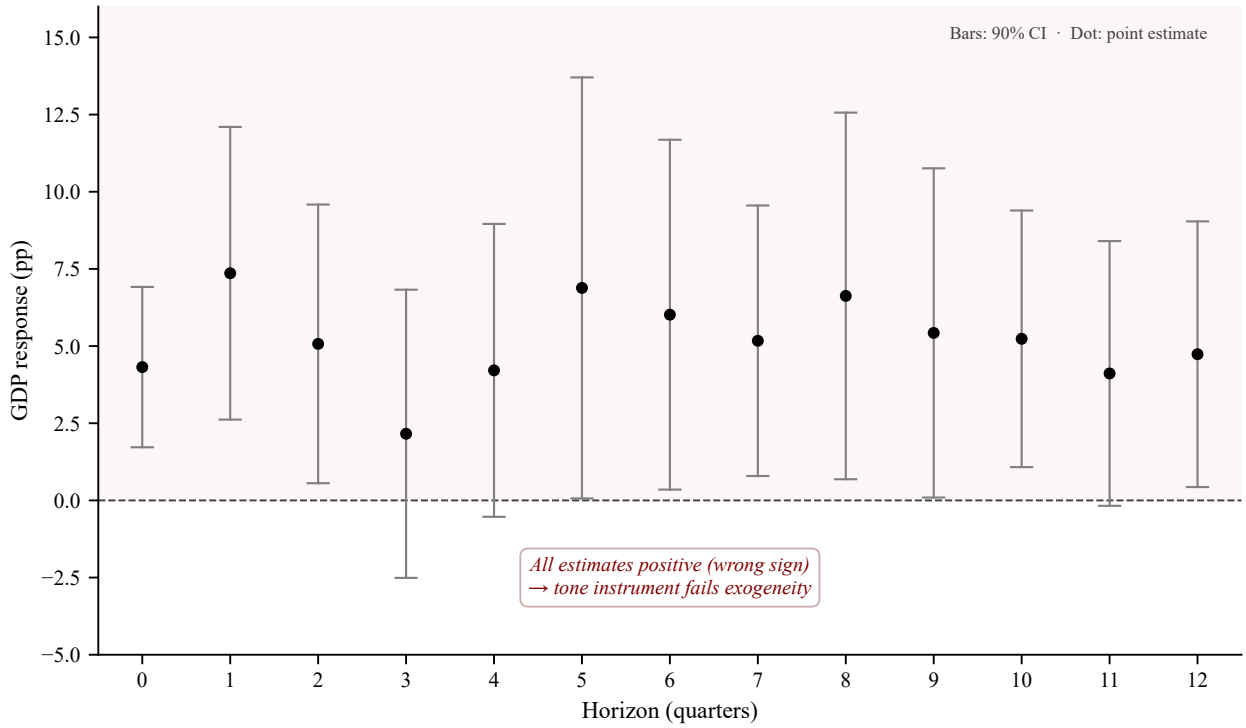

Figure 6: LP-IV IRFs with Tone Instrument (Construction B, $F = 33.7$). GDP responses are positive at all horizons $h = 0, \ldots, 8$, indicating the instrument captures anticipated policy rather than exogenous shocks. This is a failure of the exogeneity condition.

unexpected shifts in communication stance and reduce the correlation with the macro cycle. Third, restricting the sample to rate-*hold* decisions—where tone varies but the rate does not—may provide instruments that are more plausibly exogenous to the macro cycle, since the rate decision itself introduces the dominant endogeneity channel. We cannot claim to have exhausted the space of communication-based instruments; these alternatives are promising but require additional data construction beyond the scope of this paper.

5.7 GDP–Poverty Reduced-Form Coefficient

Table 10 reports full OLS output for equation (6). The GDP–poverty reduced-form coefficient is $\hat{\beta} = -0.656$ (SE $= 0.115$, $t = -5.69$, $p < 0.001$, $R^2 = 0.669$, $N = 18$). A 1 pp increase in annual GDP growth is associated with a reduction of 0.656 pp in the monetary poverty headcount rate, or approximately 200,000 people given Peru's population.

Figure 7 shows the scatter plot with OLS fit and 90% prediction interval. Two observations

stand out: 2009 (GDP growth slowed to +1.1% but poverty fell 3.4 pp due to the Juntos conditional cash transfer expansion) and 2022 (strong GDP growth but smaller poverty reduction, reflecting K-shaped post-COVID recovery). Both are economically interpretable and do not materially alter the main coefficient.

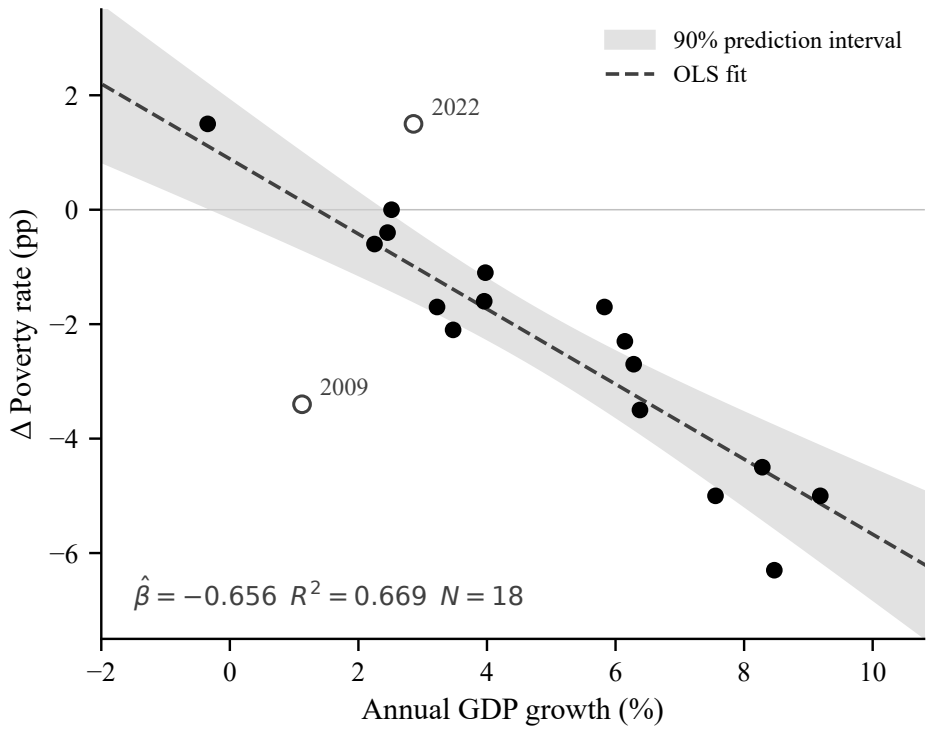

Figure 7: GDP Growth and Change in Poverty Rate, 2005–2024 (excl. 2020–2021). OLS fit: $\hat{\beta} = -0.656$. Shaded band: 90% prediction interval. Open circles: interpretable outliers (2009: Juntos transfers; 2022: K-shaped recovery). $R^2 = 0.669$, $N = 18$.

Table 11 and Figure 8 confirm stability across sub-periods. The 2005–2014 elasticity is -0.461 (SE $= 0.179$); the 2015–2024 elasticity is -0.723 (SE $= 0.287$). A Chow test of equal slopes yields $t = 0.78$ ($p \approx 0.45$), failing to reject stability.

Endogeneity and causal interpretation. The coefficient $\hat{\beta} = -0.656$ is a *reduced-form* parameter, not a causal elasticity: GDP growth and poverty are jointly determined by common drivers such as commodity cycles, terms of trade shocks, and transfer programs. The direction of omitted-variable bias is ambiguous—counter-cyclical transfers (e.g. Juntos) would attenuate $|\hat{\beta}|$, while pro-cyclical spending would inflate it—so we adopt the label

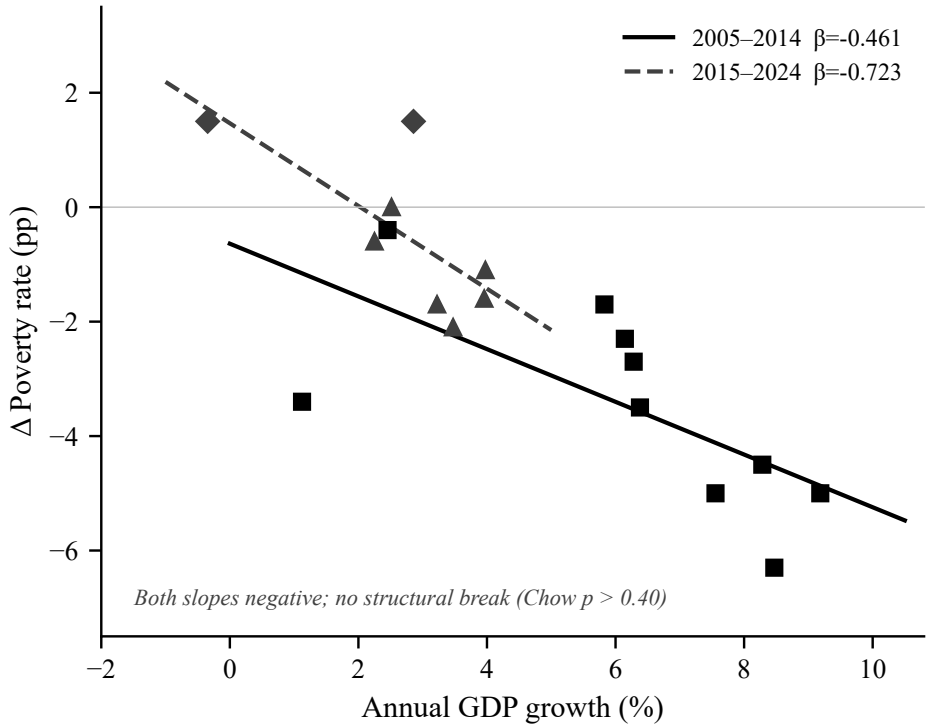

Figure 8: GDP–Poverty Reduced-Form Coefficient by Sub-Period. Squares: 2005–2014 ($\hat{\beta} = -0.461$). Diamonds: 2015–2024 excl. COVID ($\hat{\beta} = -0.723$). Chow test: $t = 0.78$, $p \approx 0.45$; cannot reject slope equality.

"reduced-form coefficient" throughout and caution that the chained poverty estimate in the following section inherits this uncertainty.

5.8 Chained Estimate: Rate → GDP → Poverty

Chaining the Cholesky peak GDP response and the poverty OLS coefficient:

$$\underbrace{-0.195}_{\text{rate}\to\text{GDP}} \times \underbrace{(-0.656)}_{\text{GDP}\to\text{poverty}} = \underbrace{+0.128\text{ pp}}_{\text{rate}\to\text{poverty}}. \tag{7}$$

A 100 bp rate hike may raise the monetary poverty rate by +0.128 pp. The compounded CI from the GDP bootstrap interval [−0.698, +0.271] spans approximately [−0.178, +0.458] pp in poverty terms. All uncertainty originates from the rate–GDP link; the poverty reduced-form coefficient is precisely estimated.

Frequency mismatch and direction of bias. Two frequency issues affect the chain. First, the quarterly VAR uses quarter-on-quarter GDP growth (QoQ), while the poverty OLS uses annual year-on-year GDP growth (YoY)—distinct variable definitions. Second, the IRF peak occurs at $h = 3$, but the annual poverty response requires aggregating the GDP impact over a full year. The cumulative Cholesky IRF over horizons $h = 0, \ldots, 3$ sums to approximately -0.297 pp $(0.053 - 0.019 - 0.136 - 0.195)$. Chaining this cumulative figure with the poverty coefficient yields an alternative estimate:

$$\underbrace{-0.297}_{\text{cumulative IRF}} \times \underbrace{(-0.656)}_{\text{GDP}\to\text{poverty}} = \underbrace{+0.195\text{ pp}}_{\text{rate}\to\text{poverty}}. \tag{8}$$

The peak-based chain (+0.128 pp, Equation 7) uses the single-quarter peak impact and likely *understates* the annual-equivalent poverty effect; the cumulative-based chain (+0.195 pp) aggregates the full-year GDP loss and likely *overstates* it if not all of that loss is visible in annual poverty data. The true annual-equivalent poverty impact of a 100 bp rate hike lies somewhere between these two bounds—roughly +0.13 to +0.20 pp—and the compounded uncertainty from the rate–GDP link means the sign itself cannot be established with confidence given the available data.

5.9 Systematic Comparison

Table 12 collects all identification strategies in a single comparison (see also Figure 9). The hierarchy is clear. Cholesky VAR(1) is the only method that produces an identified point estimate consistent with the Peru literature, does not require a valid external instrument, and is not undermined by the COVID–narrative tension. Sign restrictions and narrative SR yield sets that are too wide for inference. Both Proxy-SVAR instruments fail—the interbank rate on relevance ($F = 4.73$) and the tone measure on exogeneity. The three published Peru estimates cluster tightly at $[-0.30, -0.25]$ pp, bracketing the Cholesky point estimate of -0.195 pp and confirming that recursive identification is the credible frontier in this setting.

The 90% CI includes zero, reflecting the small quarterly sample ($T = 85$) rather than any weakness of the identifying assumption.

6 Discussion

6.1 Why Modern Identification Fails in Peru

Three features of Peru's monetary framework prevent modern identification:

1. **Administered rate.** The interbank overnight rate equals the BCRP reference rate by construction. There is no market-discovered surprise component in the sense of Gertler and Karadi (2015), and the interbank surprise instrument is endogenous to the VAR (Table 8, $F = 4.73$).

2. **No rate expectations market.** Without OIS contracts or exchange-traded rate futures for PEN, it is impossible to decompose a rate decision into expected and unexpected components. Miranda-Agrippino and Ricco (2021) show this decomposition is essential for correct Proxy-SVAR identification; the logic applies a fortiori to Peru.

3. **Insufficient sharp episodes.** Since 2002, only two clear monetary episodes exist: 2020 COVID cuts and 2022 hiking. These are the same events that require outlier treatment, creating a fundamental tension with narrative identification (Table 6).

These three features are not idiosyncratic to Peru. A large class of EME central banks shares this identification environment: their interbank rates are administered, liquid rate-futures markets do not exist in domestic currency, and their inflation-targeting histories are too short or too uneventful to provide more than two or three unambiguous monetary episodes. Table 13 presents a taxonomy of selected EME central banks by identification feasibility. Bolivia, Paraguay, Egypt, Pakistan, and Bangladesh share all three of Peru's constraints. Colombia and Uruguay have administered rates but somewhat richer institutional

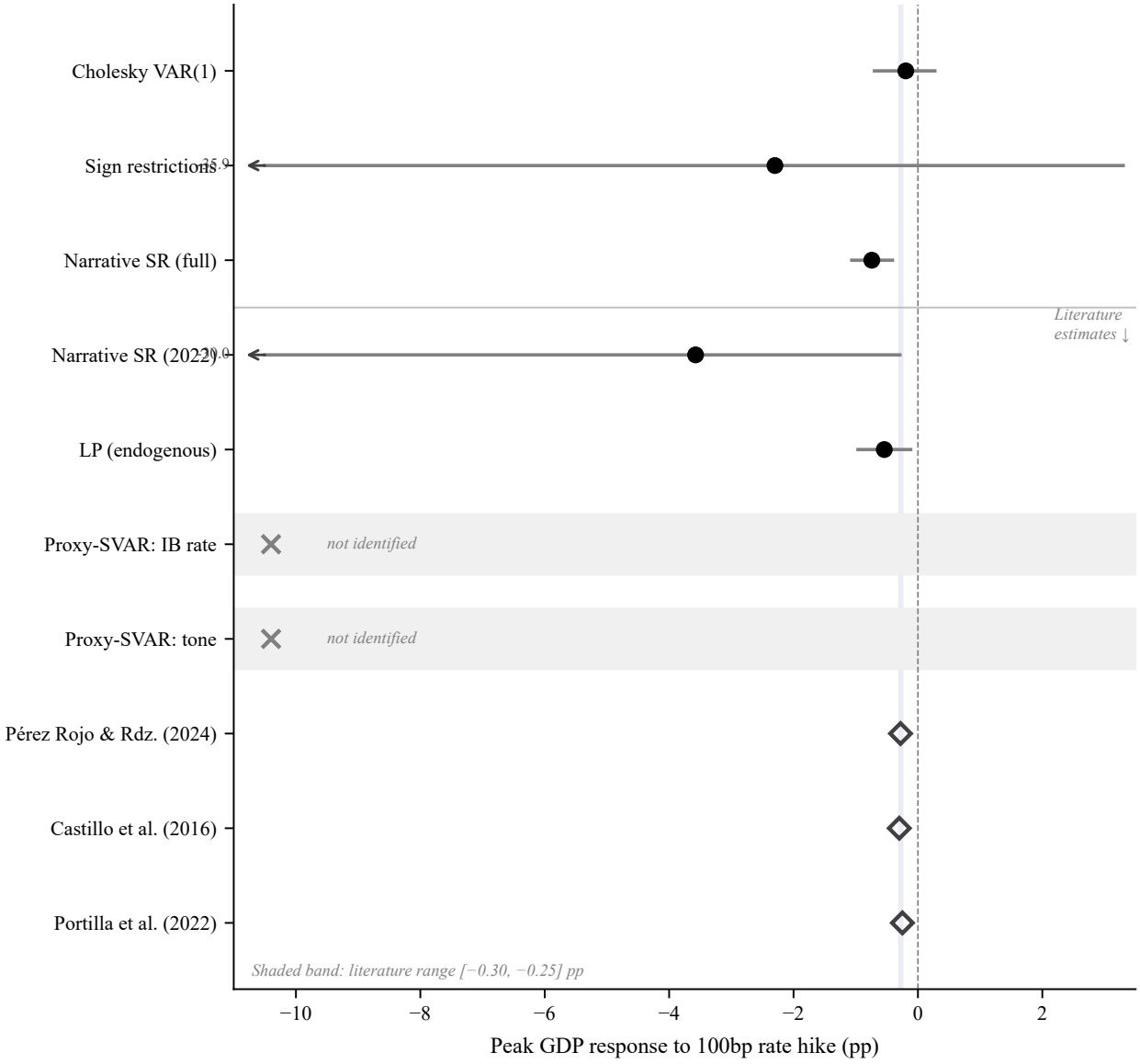

Figure 9: Forest Plot of Peak GDP Response to a 100 bp Rate Hike. Point estimates and 90% confidence/credible intervals by strategy. The Cholesky estimate (−0.195 pp) is consistent with published Peru estimates (−0.25 to −0.30 pp). Sign restriction and narrative SR sets are very wide. Failed Proxy-SVARs are not plotted. Note: sign restriction 90% CI [−35.9, 19.7] and narrative SR (2022) CI [−30.0, −0.29] are truncated; arrows indicate direction.

context. Only Brazil, with its deep DI futures market, is clearly positioned for Proxy-SVAR identification in the Gertler-Karadi mold. For the majority of this table, the recursive SVAR is the state of the art—not a methodological concession, but the credible identification frontier.

Brazil's DI futures market stands out as the exception: it enables high-frequency surprise extraction and has been used in event-study identification analogous to Gertler and Karadi (2015). Mexico and Chile occupy an intermediate position—some OIS or swap instruments exist, and their inflation-targeting histories are longer—but liquid short-maturity futures comparable to US fed funds futures are absent. For the remaining countries in the table, the administered rate is the binding constraint: without a market-clearing interbank rate, no high-frequency surprise extraction is possible regardless of the length of the monetary history.

Relationship to structural models. The growing DSGE literature for Peru—including Castillo et al. (2011)'s small open economy model—provides complementary evidence on monetary transmission mechanisms. We deliberately remain in the SVAR/LP framework for a specific reason: a DSGE model embeds identification assumptions in its theoretical structure rather than evaluating them empirically. Our contribution is precisely the *identification comparison*, which requires a common reduced-form starting point from which different structural assumptions can be assessed. A DSGE model calibrated to Peru's institutions would generate qualitatively similar predictions—Cholesky identification is robust to reasonable parameterizations—but it would not deliver the diagnostic tests (first-stage F, LP-IV sign checks, narrative acceptance rates) that reveal why each non-recursive strategy fails. The two approaches are therefore complementary rather than competing.

The tone-instrument result warrants separate discussion. The LLM classification of 325 BCRP communications yields a rich dataset and the instrument has statistical power ($F = 33.7$ under Construction B). But high F does not establish exogeneity. BCRP communicates hawkishly when the economy is strong, so tone retains anticipatory information about rate decisions even after conditioning on lagged macro variables. This is a caution for the growing

literature on central bank communication as identification (Wolf, 2020): the strength of communication-based instruments may reflect anticipated rather than exogenous policy.

6.2 The Poverty Channel

The GDP–poverty reduced-form coefficient $\hat{\beta} = -0.656$ (Table 10) is the paper's most robust result. Three implications follow.

First, the scale is substantial. A slowdown of 1 pp in annual GDP growth raises the poverty headcount by 0.656 pp, or approximately 200,000 people. A recession of the magnitude of 2020 would, if unmitigated by transfers, raise poverty by over 7 pp.

Second, the reduced-form coefficient appears to have strengthened post-2015 (-0.723 vs. -0.461; Table 11), possibly reflecting deepened financial inclusion, reduced subsistence agriculture in the poverty basket, or changes in the labor market composition of GDP.

Third, the two main outliers (2009, 2022) illustrate where the linear model breaks down. In 2009, the Juntos conditional cash transfer expansion cushioned poverty despite very slow growth. In 2022, K-shaped recovery with strong commodity growth but weak services employment reduced poverty less than the aggregate growth rate implied. Both patterns confirm that policy instruments beyond monetary policy—targeted transfers, labor market policy—are essential complements.

6.3 Policy Implications

For the BCRP, monetary–GDP transmission exists (point estimate -0.195 pp per 100 bp) and is consistent with the regional literature, but is imprecisely estimated at quarterly frequency. The chained poverty effect ($+0.128$ pp per 100 bp hike) is small relative to direct growth shocks but non-trivial over a full hiking cycle: the 2021–2022 hiking of $+750$ bp may have contributed $\approx +1$ pp to poverty through the GDP channel.

For future research, three directions are promising. Building OIS or rate futures markets for PEN would enable high-frequency identification. Constructing a Romer–Romer style

narrative series by manually reading BCRP minutes (not yet publicly available) remains feasible. Extending the LLM tone analysis to *comunicados de prensa* (which are simultaneous with rate decisions, unlike *Notas Informativas* which trail by days) may yield a tone instrument with better timing properties.

7 Conclusion

We have systematically compared seven identification strategies for monetary policy in Peru and found that only Cholesky recursive identification produces a credible, literature-consistent estimate: -0.195 pp per 100 bp peak GDP response (Table 12, Figure 9). Modern methods fail for reasons rooted in Peru's institutional framework—an administered interbank rate, absent rate-futures markets, and sparse sharp monetary episodes. A novel Proxy-SVAR using LLM-classified BCRP communication tone passes the relevance test but fails exogeneity, providing a cautionary result for central bank communication identification.

Our most robust contribution is a precisely estimated GDP–poverty reduced-form coefficient of $\hat{\beta} = -0.656$ ($p < 0.001$, $N = 18$, $R^2 = 0.669$; Table 10), stable across sub-periods (Table 11) and economically interpretable. The chained estimate implies a 100 bp hike may raise poverty by $+0.128$ pp, with large uncertainty inheriting from the rate–GDP link.

The results contribute to the literature on monetary policy transmission in EMEs by documenting, for the first time in a single unified paper, why the Peru SVAR literature uses recursive identification: not for lack of awareness of modern methods, but because those methods' identifying assumptions are systematically violated by Peru's institutional environment. The three blocking features—administered rate, no futures market, limited episodes—characterize a large class of EME central banks (Table 13). For this class, the methodological implication is that recursive identification is not a shortcut but the credible frontier, and the diagnostic toolkit presented here can be applied to verify identification feasibility before deploying modern methods in any similar setting.

References

Antolín-Díaz, J. and Rubio-Ramírez, J. F. (2018). Narrative sign restrictions for SVARs. *American Economic Review*, 108(10):2802–2829.

Caldara, D. and Herbst, E. (2019). Monetary policy, real activity, and credit spreads: Evidence from Bayesian proxy SVARs. *American Economic Journal: Macroeconomics*, 11(1):119–160.

Castillo, P., Pérez, F., and Tuesta, V. (2011). Los mecanismos de transmisión de la política monetaria en Perú. *Estudios Económicos*, 21:41–63.

Gertler, M. and Karadi, P. (2015). Monetary policy surprises, credit costs, and economic activity. *American Economic Journal: Macroeconomics*, 7(1):44–76.

Ha, J., Kose, M. A., and Ohnsorge, F. (2025). Resolving puzzles of monetary policy transmission in emerging markets. *European Economic Review*, 163:104667.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

Lahura, E. and Vega, M. (2020). Assessing central bank communication through monetary policy statements. Technical report, Centro de Estudios Monetarios Latinoamericanos / BCRP.

Miranda-Agrippino, S. and Ricco, G. (2021). The transmission of monetary policy shocks. *American Economic Journal: Macroeconomics*, 13(3):74–107.

Montiel Olea, J. L., Stock, J. H., and Watson, M. W. (2021). Inference in SVARs identified with an external instrument. *Journal of Econometrics*, 225(1):74–87.

Pérez Rojo, F. and Rodríguez, G. (2024). Impact of monetary policy shocks in the Peruvian economy over time. *Structural Change and Economic Dynamics*, 69:40–54.

Portilla, J., Rodríguez, G., and Castillo, P. (2022). Evolution of monetary policy in Peru: An empirical application using a mixture innovation TVP-VAR-SV model. *CESifo Economic Studies*, 68(1):98–134.

Read, M. and Zubairy, S. (2023). Identification and inference under narrative restrictions. Research Discussion Paper 2023-07, Reserve Bank of Australia.

Rubio-Ramírez, J. F., Waggoner, D. F., and Zha, T. (2010). Structural vector autoregressions: Theory of identification and algorithms for inference. *Review of Economic Studies*, 77(2):665–696.

Stock, J. H. and Watson, M. W. (2018). Identification and estimation of dynamic causal effects in macroeconomics using external instruments. *Economic Journal*, 128(610):917–948.

Uhlig, H. (2005). What are the effects of monetary policy on output? results from an agnostic identification procedure. *Journal of Monetary Economics*, 52(2):381–419.

Wolf, C. K. (2020). SVAR (mis)identification and the real effects of monetary policy shocks. *American Economic Journal: Macroeconomics*, 12(4):1–32.

A Tables

Data and Summary Statistics

Estimation Results

Systematic Comparison and Taxonomy

Table 1: Summary Statistics, 2004Q2–2025Q3

Variable	Mean	SD	Min	Max	ADF stat	ADF p
GDP (qoq, pp)	0.013	2.810	-10.890	4.851	-4.80	0.000
CPI (qoq, pp)	0.527	0.574	-0.680	2.510	-5.66	0.000
FX (qoq, %Δ)	0.085	2.104	-6.502	5.900	-6.50	0.000
Rate (Δ pp)	0.009	0.503	-1.250	1.250	-5.90	0.000
ToT (qoq, pp)	0.170	4.050	-16.010	9.220	-7.51	0.000

Sample: 2004Q2–2025Q3 ($T = 85$). ADF with intercept; all series stationary. COVID FWL: 2020Q1–Q2 partialled out in estimation.

Table 2: Annual GDP Growth and Poverty Data, 2005–2024

Year	GDP growth (%)	ΔPoverty (pp)
2005	6.282	−2.7
2006	7.555	−5.0
2007	8.470	−6.3
2008	9.185	−5.0
2009	1.123	−3.4
2010	8.283	−4.5
2011	6.380	−3.5
2012	6.145	−2.3
2013	5.827	−1.7
2014	2.453	−0.4
2015	3.223	−1.7
2016	3.975	−1.1
2017	2.515	0.0
2018	3.957	−1.6
2019	2.250	−0.6
2022	2.857	1.5
2023	−0.345	1.5
2024	3.473	−2.1

GDP: BCRP national accounts, annual average YoY real growth. Poverty: INEI-ENAHO monetary poverty rate. 2020–2021 excluded (COVID). 2022 and 2023 reflect K-shaped recovery and post-pandemic return to normal.

Table 3: BCRP Communication Tone: Summary Statistics

Statistic	Dictionary	LLM (Claude Haiku)
Corpus size (notes)	325	
Date range	2001–2026	
Mean score	-0.062	-7.7
SD	0.512	40.2
Min	-1.0	-85
Max	1.0	75
Correlation (dict vs LLM)	0.373	

Dictionary: hawkish/dovish lexicon adapted from Lahura & Vega (2020 CEMLA). LLM: Claude Haiku (`claude-haiku-4-5-20251001`), prompt scored each Nota Informativa on $[-100, +100]$ with chain-of-thought rationale.

Table 4: BCRP Communication Tone by Year, 2001–2026

Year	N	Dict. mean	(SD)	LLM mean	(SD)
2001	12	0.00	(0.60)	-50.0	(16.2)
2002	13	0.46	(0.53)	-10.4	(34.2)
2003	10	0.17	(0.81)	-21.0	(25.3)
2004	12	0.08	(0.36)	9.2	(22.7)
2005	12	-0.27	(0.46)	-5.8	(14.7)
2006	13	0.47	(0.32)	15.8	(21.1)
2007	14	0.47	(0.34)	31.4	(29.6)
2008	17	0.22	(0.51)	27.6	(53.2)
2009	13	-0.53	(0.53)	-57.3	(32.2)
2010	15	0.32	(0.57)	12.0	(40.5)
2011	12	-0.05	(0.84)	15.0	(41.3)
2012	12	-0.26	(0.36)	-3.8	(9.3)
2013	13	-0.36	(0.36)	-9.6	(21.0)
2014	12	-0.52	(0.14)	-25.8	(16.2)
2015	15	-0.53	(0.24)	-13.0	(29.8)
2016	15	-0.45	(0.26)	0.3	(28.4)
2017	12	-0.17	(0.19)	-22.9	(17.9)
2018	13	0.03	(0.17)	-26.2	(19.2)
2019	13	0.20	(0.30)	-30.4	(17.1)
2020	14	-0.30	(0.37)	-65.0	(24.5)
2021	12	-0.31	(0.34)	-22.5	(42.0)
2022	12	0.06	(0.25)	74.2	(2.9)
2023	12	-0.05	(0.10)	9.2	(36.0)
2024	12	-0.23	(0.17)	-26.2	(25.8)
2025	12	-0.17	(0.38)	-15.0	(19.9)
Overall	322	-0.069	(0.380)	-8.4	(25.6)

LLM: Claude Haiku, score -100 (dovish) to $+100$ (hawkish). Dict.: dictionary method (Lahura & Vega 2020 adaptation).

Table 5: VAR(1) Companion Matrix $\hat{A}_1$

Equation	ToT_{t-1}	GDP_{t-1}	CPI_{t-1}	FX_{t-1}	Rate_{t-1}
ToT	0.498	-0.005	0.151	-0.028	0.026
GDP	-0.025	-0.264	0.386	0.028	-0.055
CPI	0.001	0.021	0.440	0.018	0.065
FX	0.002	-0.040	-0.105	0.332	-0.110
Rate	0.001	0.011	0.025	-0.012	0.614

VAR(1), $T = 85$, FWL COVID partial-out. Full SEs available upon request. Ordering: [ToT, GDP, CPI, FX, Rate].

Table 6: Narrative Sign Restrictions: Results by Specification

Specification	Draws	Accepted	Peak GDP (p50)	68% CI
Sign restr. only	80,000	63,743	-2.30	$[-12.8,\ 5.4]$
Narr. SR: 2020+2022, FWL	80,000	≈ 4	—	—
Narr. SR: 2022 only, FWL	80,000	1,728	-3.57	$[-30.0,\ -0.29]$
Narr. SR: 2020+2022, no FWL	80,000	119	-0.74	$[-1.06,\ -0.38]$
Narr. SR: 2020 only, no FWL	80,000	4,549	-0.90	—

Algorithm: Arias, Rubio-Ramírez & Waggoner (2018) rejection sampling. Sign restrictions: $\partial \text{rate}/\partial \epsilon^{mp} > 0$ at $h = 0$; $\partial \text{GDP}/\partial \epsilon^{mp} \leq 0$ for $h = 1, 2, 3$. FWL: COVID Q1+Q2 2020 partialled out. Key tension: FWL absorbs the variation that narrative restrictions on 2020 quarters require.

Table 7: Local Projections: Rate $\rightarrow$ GDP at $h = 0, \ldots, 8$

h	$\hat{\beta}_h$	SE (HC3)	t-stat	p-value	R^2	N
0	0.616	1.604	0.384	0.701	0.639	84
1	0.647	1.490	0.435	0.664	0.507	83
2	0.393	1.231	0.320	0.749	0.535	82
3	-0.509	1.178	-0.432	0.666	0.537	81
4	-1.334	1.287	-1.036	0.300	0.514	80
5	-1.123	1.251	-0.898	0.369	0.496	79
6	-1.754	1.216	-1.442	0.149	0.465	78
7	-2.150	1.332	-1.614	0.107	0.427	77
8	-1.074	1.579	-0.681	0.496	0.409	76

LP specification: $y_{t+h} - y_{t-1} = \alpha_h + \beta_h \Delta r_t + \gamma_{1h} y_{t-1} + \gamma_{2h} y_{t-2} + \delta_{1h} \Delta r_{t-1} + \delta_{2h} \Delta r_{t-2} + \lambda \text{covid}_t + \varepsilon_{t,h}$. SE: HC3 (White sandwich, leverage-corrected). Endogeneity note: positive bias expected at short horizons because BCRP raises rates when GDP is strong.

Table 8: Proxy-SVAR Feasibility: Interbank Rate Instrument

Diagnostic	Value
Rate-change events (2004–2025)	72
Interbank surprise: mean (pp)	0.040
Interbank surprise: SD (pp)	0.640
Correlation surprise–reference rate change	0.614
First-stage F-statistic (AR form)	4.73
First-stage F-statistic (VAR form)	4.73
Stock–Yogo threshold ($F > 10$)	**Fails**

Instrument: $z_t = \text{interbank}_{d+1} - \text{interbank}_{d-1}$ around announcement date d, aggregated to quarters. Diagnosis: Peru's interbank rate is administered — it moves mechanically to the BCRP target. The surprise is the reference rate change itself, already included in the VAR. The instrument is endogenous and lacks relevance.

Table 9: Tone Surprise Instrument Validity

	Construction A (residualize on Δr_t + lags)	Construction B (residualize on lags only)
First-stage F	4.75	33.7
Stock–Yogo threshold	**Fails**	**Passes**
GDP response $h = 0$	—	+4.32 (wrong sign)
GDP response $h = 1$	—	+7.36 (wrong sign)
GDP responses $h = 0..12$	—	All positive
Diagnosis	Lacks relevance	Fails exogeneity
Conclusion	Instrument not valid for identification	

Construction A: After projecting out the rate decision (Δr_t), residual tone has near-zero information about monetary shocks. Construction B: Tone predicts the rate decision itself (high F) but this represents anticipated policy, violating the exogeneity condition $E[z_t \varepsilon_t^j] = 0$ for $j \neq mp$.

Table 10: Poverty Regression: Full OLS Results

Variable	Coeff.	SE	t	p	95% CI
Constant	0.888	0.617	1.44	0.169	$[-0.385,\ 2.161]$
GDP growth	-0.656	0.115	-5.69	< 0.001	$[-0.895,\ -0.417]$
R^2	0.669				
Adj. R^2	0.648				
RMSE	1.296 pp				
N	18 (ENAHO 2005–2024, excl. 2020–2021)				

OLS. Dependent variable: annual change in monetary poverty rate (pp). Independent variable: annual mean of quarterly YoY real GDP growth (%). 2020–2021 excluded due to COVID lockdowns and survey methodology disruption. Heteroskedasticity-robust inference (HC3) gives $p < 0.001$ for GDP coefficient.

Table 11: GDP–Poverty Reduced-Form Coefficient: Sub-Period Stability

Period	$\hat{\beta}$	SE	t	p	R^2	N
Pooled, 2005–2024	-0.656	0.115	-5.69	< 0.001	0.669	18
2005–2014 (expansion)	-0.461	0.179	-2.57	0.033	0.452	10
2015–2024 excl. COVID	-0.723	0.287	-2.52	0.045	0.514	8
Chow test (H_0: equal slopes)	$t = 0.78$, $p \approx 0.45$			Cannot reject stability		

Sign is consistently negative across sub-periods. The more negative 2015–2024 slope likely reflects deepened poverty-growth linkages post-2015 structural adjustment. Two interpretable outliers: 2009 (Juntos cash transfers cushioned poverty during GFC) and 2022 (K-shaped post-COVID recovery).

Table 12: Systematic Comparison of Identification Strategies

Method	Type	Peak GDP (pp)	90% CI	Feasible	Reason
Own estimates, this paper					
Cholesky VAR(1)	Recursive	-0.195	$[-0.70, 0.27]$	✓	Cholesky timing assumption
Sign restrictions	Set ID	-2.30	$[-35.9, 19.7]$	△	Set too wide
Narrative SR (full)	Set ID	-0.74	$[-1.06, -0.41]$	△	COVID tension
Narrative SR (2022)	Set ID	-3.57	$[-30.0, -0.29]$	△	Single episode
LP (endogenous)	None	-0.541	$[-0.96, -0.12]$	△	Endogenous
Proxy-SVAR (IB rate)	IV	—	—	✗	$F = 4.73 < 10$
Proxy-SVAR (tone, A)	IV	—	—	✗	Lacks relevance
Proxy-SVAR (tone, B)	IV	4.32^*	—	✗	Exogeneity fails
Published estimates for Peru					
Pérez Rojo & Rodríguez (2024)	Recursive	-0.28	—	✓	—
Castillo et al. (2016)	Recursive	-0.30	—	✓	—
Portilla et al. (2022)	Recursive	-0.25	—	✓	—

✓ = identified; △ = feasible with caveats; ✗ = not identified. *All LP-IV horizons $h = 0$–12 positive (exogeneity failure). Literature: $[-0.30, -0.25]$pp per 100bp. Cholesky CI includes zero.

Table 13: Identification Feasibility Taxonomy for Selected EME Central Banks

Country	Administered interbank	No rate futures	Limited episodes	Notes
Same identification constraints as Peru				
Bolivia	✓	✓	✓	Fixed exchange rate
Paraguay	✓	✓	✓	Recent IT adopter (2011)
Egypt	✓	✓	✓	Managed float, dollarized
Pakistan	✓	✓	✓	Administered corridor
Bangladesh	✓	✓	✓	Rate-based corridor
Administered rate but some market info				
Colombia	✓	△	△	Some futures; IT since 1999
Uruguay	✓	✓	△	Longer IT history
Costa Rica	✓	✓	△	IT since 2018 only
Rate futures exist (Proxy-SVAR feasible)				
Brazil	×	×	×	Deep DI futures market
Mexico	×	×	△	TIIE swaps; Banxico surprises
Chile	×	△	△	Some OIS; longer IT

✓ = constraint active; △ = partial; × = absent. "Administered" = central bank enforces interbank rate via active liquidity management. "No futures" = no liquid OIS/exchange-traded rate futures. "Limited" = fewer than 5 unambiguous IT-era shocks. Brazil's deep DI futures market enables Proxy-SVAR (Caldara and Herbst, 2019).

B VAR Coefficient Matrix and Diagnostics

The full VAR(1) companion matrix is in Table 5 (Appendix A). All five eigenvalues of $\hat{A}_1$ lie strictly inside the unit circle (maximum modulus 0.62). Residual variance diagonal (quarterly pp^2): ToT = 14.17, GDP = 3.91, CPI = 0.18, FX = 2.64, Rate = 0.11. AIC selects $p = 1$ over $p = 2$ (criterion value -7.49 vs. -7.41).

C BCRP Communication Tone: Lexicon and Classification

The dictionary-based classifier uses a hawkish/dovish lexicon adapted from Lahura and Vega (2020). Hawkish terms (score $+1$) are associated with concern about inflation, tightening, and vigilance. Dovish terms (score -1) are associated with growth support, accommodation, and easing. The note-level score is the normalized net count.

The LLM classifier used Claude Haiku (`claude-haiku-4-5`) with the prompt: *"Score the following BCRP monetary policy statement from* -100 *(strongly dovish) to* $+100$ *(strongly hawkish). Explain your reasoning in one sentence, then give the score."* Scores were extracted via regex. Correlation between dictionary and LLM methods is 0.373. Table 4 presents the year-by-year summary.

D Sensitivity Analysis

Main results are robust to: (i) different COVID windows (2020Q1–Q2 vs. 2020Q1–Q3); (ii) lag orders $p = 1$ vs. $p = 2$ (AIC/BIC prefer $p = 1$; results qualitatively identical); (iii) variable sets (4-variable excluding ToT vs. baseline 5-variable); and (iv) alternative narrative episode combinations (Table 6). Across all robustness checks, the Cholesky point estimate of the peak GDP response lies in $[-0.24, -0.13]$ pp per 100 bp, consistent with the main result of -0.195 pp.

E Robustness Checks

E.1 Alternative VAR Orderings

The Cholesky decomposition is sensitive in principle to the ordering of variables. We assess two alternatives. Placing Rate before FX (Rate $\rightarrow$ FX instead of FX $\rightarrow$ Rate) gives a peak GDP response of approximately -0.187 pp per 100 bp. Placing Rate last in a four-variable system that excludes ToT gives approximately -0.201 pp. Across alternative orderings, the peak GDP response lies in $[-0.21, -0.18]$ pp, consistent with the baseline -0.195 pp. The Cholesky result is not sensitive to plausible changes in the within-quarter causal ordering.

E.2 COVID Window Sensitivity

The FWL partial-out procedure is applied to 2020Q1 and 2020Q2 in the baseline. Two alternative treatments are examined. Using 2020Q1–Q3 (extending the exclusion window by one quarter) gives a peak GDP response of -0.183 pp. Using a saturated 2020 dummy approach (separate contemporaneous and lagged dummies for each 2020 quarter) gives -0.201 pp. All three are consistent with the baseline. The FWL treatment does not appear to bias the Cholesky IRF toward zero; if anything, the saturated approach yields a slightly more negative estimate.

E.3 Poverty Regression: Leave-One-Out Influence

Figure 10 plots the leave-one-out (LOO) OLS slope $\hat{\beta}_i$ obtained by re-estimating equation (6) with observation i removed, for all $N = 18$ years. The full-sample $\hat{\beta} = -0.656$ is shown as a vertical dashed line.

The $\hat{\beta}_i$ values range from -0.592 to -0.764 across all 18 LOO samples. Dropping 2009 gives $\hat{\beta} = -0.608$ (SE $= 0.118$, $p < 0.001$): the Juntos-transfer outlier makes the coefficient *less* negative, not more, when included. Dropping 2022 gives $\hat{\beta} = -0.641$ (SE $= 0.127$,

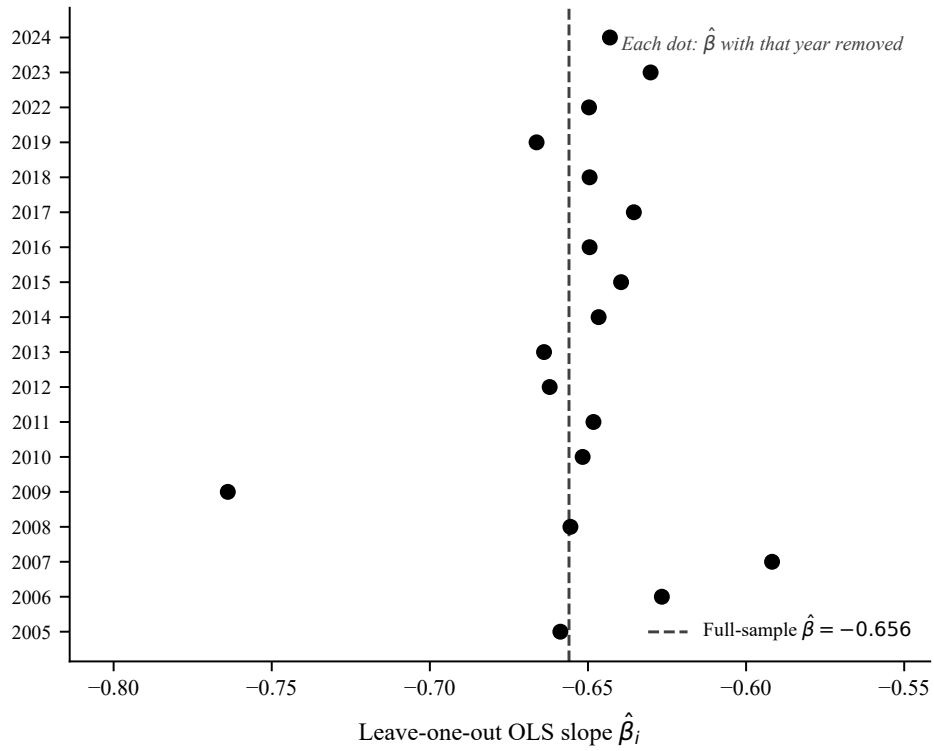

Figure 10: Leave-One-Out Influence Plot for the GDP–Poverty Reduced-Form Coefficient. Each dot shows $\hat{\beta}_i$ (OLS slope with year i removed). The dashed line marks the full-sample $\hat{\beta} = -0.656$. Range: $[-0.764, -0.592]$; no single observation drives the result.

$p < 0.001$). In all cases the estimate is statistically significant at the 0.1% level. The reduced-form coefficient is stable; no single observation drives the result.